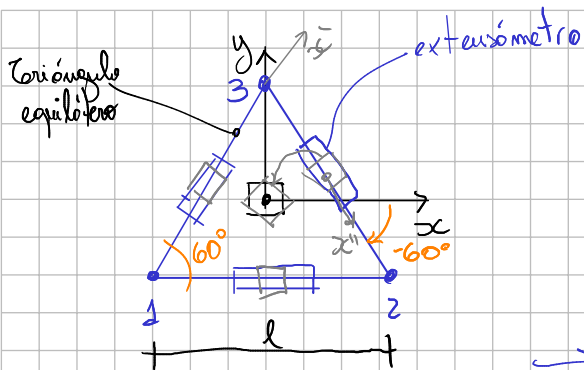
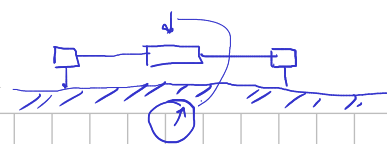
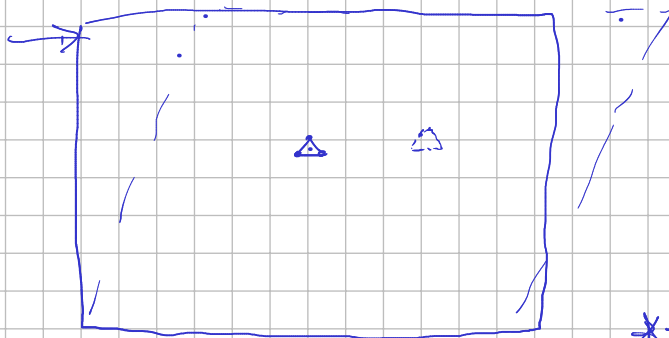


Pequeñas deformaciones



$$\Delta l_{12}, \Delta l_{23}, \Delta l_{31} \rightarrow \text{Datos}$$

$$\epsilon_{xx} = \frac{ds_x - ds_{ox}}{ds_{ox}} = \frac{\Delta l_{12}}{l_{12}} = \frac{\Delta l_{12}}{l}$$



$$\begin{cases} * \epsilon_{xx} = \frac{\Delta l_{12}}{l} \\ \epsilon_{x'x'} = \frac{\Delta l_{13}}{l} \\ \epsilon_{x''x''} = \frac{\Delta l_{23}}{l} \end{cases}$$

$$\epsilon = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} \\ \epsilon_{xy} & \epsilon_{yy} \end{bmatrix}$$

* $\rightarrow$ 3 ecuaciones
 $\downarrow$
3 incógnitas
($\epsilon_{xx}, \epsilon_{yy}, \epsilon_{xy}$)

$$* \epsilon_{x'x'} = \frac{\epsilon_{xx} + \epsilon_{yy}}{2} + \frac{\epsilon_{xx} - \epsilon_{yy}}{2} \cos(2\theta) + \epsilon_{xy} \sin(2\theta) \quad *1$$

$$\epsilon_{x''x''} = \frac{\epsilon_{xx} + \epsilon_{yy}}{2} + \frac{\epsilon_{xx} - \epsilon_{yy}}{2} \cos(2(-\theta)) + \epsilon_{xy} \sin(2(-\theta))$$

$$* \epsilon_{x''x''} = \frac{\epsilon_{xx} + \epsilon_{yy}}{2} + \frac{\epsilon_{xx} - \epsilon_{yy}}{2} \cos(2\theta) - \epsilon_{xy} \sin(2\theta) \quad *2$$

$$*1 + *2 \Rightarrow \epsilon_{x'x'} + \epsilon_{x''x''} = \epsilon_{xx} + \epsilon_{yy} + (\epsilon_{xx} - \epsilon_{yy}) \cos(2\theta)$$

$$= \epsilon_{xx} (1 + \cos(2\theta)) + \epsilon_{yy} (1 - \cos(2\theta))$$

$$\epsilon_{yy} = \frac{\epsilon_{x'x'} + \epsilon_{x''x''} - 2\epsilon_{xx} \cos^2\theta}{2 \sin^2\theta} \Rightarrow \epsilon_{yy} = \frac{2\Delta l_{13} + 2\Delta l_{23} - \Delta l_{12}}{3l}$$

$$*1 - *2 \Rightarrow \epsilon_{x'x'} - \epsilon_{x''x''} = 2\epsilon_{xy} \sin(2\theta) \Rightarrow \epsilon_{xy} = \frac{\epsilon_{x'x'} - \epsilon_{x''x''}}{2 \sin(2\theta)}$$

$$\epsilon_{xy} = \frac{\Delta l_{13} - \Delta l_{23}}{\sqrt{3} \cdot l}$$